

Source frames for the complex interchange

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This note moves the stage-two rank of every auxiliary role of the complex interchange into a single child, by starting the role in a nonzero phase frame. It is the complex analogue of PR #13's auxiliary source frames on the bit side.

Phase frames. For $\phi : \mathbb{F}_2^m \rightarrow \mathbb{Z}/4$ write $D_\phi = H^{\otimes m} \text{diag}(i^\phi) H^{\otimes m}$. All phase frames share the eigenbasis $H^{\otimes m}$, so $D_\psi D_\phi^{-1} = D_{\psi-\phi}$. For a subspace U let $q_U(y)$ be the weight of the orthogonal projection of y onto U modulo 4 when U has an orthonormal basis; the frame of a label U is D_{q_U} .

Lemma 1 (Auxiliary source frames). *In the common-frame form of the complex interchange, route w applies $\epsilon_w D_{\phi_{\text{sink}}(\rho w) - \phi_{\text{src}}(w)}$. An auxiliary role ($\rho(w) = w$, $\epsilon_w = +1$) may start at $D_{q_{D_0}}$ provided its sink is $D_{\text{wt}+q_{D_0}}$. Every gate, interior edge and other role is unchanged, and the network still restores arbitrary scratch.*

Proof. The common-frame identity holds for any invertible frames, so the route operator is $D_{\text{wt}+q_{D_0}-q_{D_0}} = D_{\text{wt}} = C^{\otimes m}$, the same as with source 0 and sink D_{wt} . The logical scratch is $g = D_{-q_{D_0}} f$; since f is arbitrary so is g , and the output is $D_{\text{wt}+q_{D_0}} D_{-q_{D_0}} f = C^{\otimes m} f$. $\square$

Lemma 2 (Costs). *Suppose $D_0 \subseteq U$ and $D_0 \subseteq V$, where U and V are the role's first and last stage-two labels, $U = D_0 \perp L_1$ and $V = D_0 \perp L_2$. Then the entrance edge costs $\dim L_1$ and the exit edge to the sink costs $m - \dim L_2$. The exit residual $G = D_0 \perp V^\perp$ has an orthonormal basis.*

Proof. Weights of orthogonal parts add modulo 4, because orthogonal parts overlap in an even number of coordinates. So $q_U - q_{D_0} = q_{U \cap D_0^\perp} = q_{L_1}$, of rank $\dim L_1$, and $\text{wt} + q_{D_0} - q_V = q_{V^\perp} + q_{D_0} = q_G$, of rank $m - \dim L_2$. G is an orthogonal sum of nondegenerate spaces, so it is nondegenerate. It is nonalternating because D_0 contains an odd vector ($e_j \otimes e_k \otimes t_B$ with

$j \notin A$). A nondegenerate nonalternating space over $\mathbb{F}_2$ has an orthonormal basis. $\square$

So the exit is exactly the operator of an increasing edge $0 \rightarrow G$, and PR #10's whole-residual child and its Z_B sign wrapper apply to it. Every stage-two label of the form $(B \otimes F \perp P \otimes L) \otimes Q$ contains $D_0 = B \otimes F \otimes Q$, so the hypothesis holds for every addition and piece role.

Remark (Excluded roles). Retained roles end at D_0 ; a source frame would give them an exit of rank m , which is not a contraction, so they keep source 0. Input-pivot slots meet stage two first at the copy into the data frame, which is not a frame above D_0 ; they also keep source 0. The rank sum s and the deficit are unchanged.

Effect. For each affected role the stage-two entrance child (rank $h^2 - h$) and exit child (rank $m - h^2$) merge into one child of rank $m - h$. In `independent/complex-network/fullbatch_hist.py` this is the `sf` option. For PR #7's network at $h = 28$ under full batching the certified saving rises from $1048009/(2.5 \cdot 10^{11})$ to $4079603/(2.5 \cdot 10^{11}) \approx 1.632 \cdot 10^{-5}$; every child has width at most $m - h \leq m - 1$, so the topology-free guard is unchanged.

Checks. Exact label checks at $h = 8$ and $h = 10$: the entrance and exit phase identities, G nondegenerate and nonalternating, and the ranks $m - h$ and $m - h + 1$. A full three-stage scalar simulation in the Hadamard domain modulo $2^{64} - 59$, with random gate phases, random data-endpoint phases and arbitrary scratch, passes at $h = 8$ and $h = 10$; a negative control with sink wt alone fails to restore scratch.